

$$\mathbf{e}_{t,k}^A = \mathbf{0} \quad \Rightarrow \quad \mathbf{g}_{t,k}^A = \mathbf{J}_{ZA}^\top \mathbf{e}_{t,k}^Z = \left(\frac{\partial \hat{\mathbf{Z}}_{t,k}^1}{\partial \mathbf{A}_t^{\tau_k^X}} \right)^\top \mathbf{e}_{t,k}^Z$$